Annu. Rev. Microbiol. 1993. 47:321–50

REGULATION OF THE HEAT-SHOCK RESPONSE IN BACTERIA

T. Yura[1]
Institute for Virus Research, Kyoto University, Kyoto 606-01, Japan

H. Nagai
National Institute of Genetics, Mishima, Shizuoka 411, Japan

H. Mori
Institute for Virus Research, Kyoto University, Kyoto 606-01, Japan

KEY WORDS: heat-shock protein, stress response, σ factor, translational control, *Escherichia coli*

CONTENTS

[1]Present address: HSP Research Institute, Kyoto Research Park, Kyoto 600, Japan.

0066-4227/93/1001-0321$02.00

ABSTRACT

When bacteria cells are exposed to higher temperature, a set of heat-shock proteins (hsps) is induced rapidly and transiently to cope with increased damage in proteins. The mechanism underlying induction of hsps has been a central issue in the heat-shock response and studied intensively in *Escherichia coli*. Immediately upon temperature upshift, the cellular level of σ^{32} responsible for transcription of heat-shock genes increases rapidly and transiently. This increase in σ^{32} results from both increased synthesis and stabilization of σ^{32}, which is ordinarily very unstable. A clue to further understanding of early regulatory events came from recent analysis of translational induction and subsequent shut-off of σ^{32} synthesis. Whereas a 5′-coding region of mRNA for σ^{32} is involved in the induction meditated by the mRNA secondary structure, a distinct segment of σ^{32} polypeptide further downstream is involved in the DnaK/DnaJ-mediated shut-off and destabilization of σ^{32} that may be mutually interconnected.

INTRODUCTION

When cells of bacteria or almost any organism are exposed to high temperature, the synthesis of a set of heat-shock proteins (hsps) is rapidly induced. The primary structure of most hsps appears to be highly conserved during evolution, suggesting that they serve similar functions in all organisms (27, 62). Transient induction of hsps represents an important protective and homeostatic mechanism to cope with physiological and environmental stress at the cellular level. In recent years, both the function and regulation of hsps have undergone extensive study with bacteria, yeast, and various higher organisms. Most hsps are synthesized under nonstress conditions, albeit at reduced rates, and research indicated that hsps play fundamental roles in normal cell physiology in addition to their activity under stress conditions. These proteins are often described as molecular chaperones in that they mediate the correct folding and assembly of polypeptides but are not components of the functional assembled structures (17, 76).

Induction of hsps occurs primarily at the level of transcription: transcription of heat-shock genes is enhanced as a result of increased activity of specific transcription factors—σ^{32} in bacteria such as *Escherichia coli,* and HSF (heat-shock factor) in eukaryotes such as yeast, flies, mice, and humans (62). Moreover, some hsps, particularly HSP70 (DnaK in *E. coli*), exert a negative-feedback control on the synthesis of hsps following initial induction (14, 95), presumably by interacting directly or indirectly with the heat-shock transcription factors (12). Thus not only the structure and function of hsps but their mechanisms of regulation appear to be conserved, at least in some basic forms. Such a remarkable conservation must reflect the rapidly changing cellular requirements for hsps of most living organisms in nature and the need for rapid and fine adjustment of hsp levels to assure optimal growth and survival.

One of the outstanding problems in the field of heat-shock and stress response has been to elucidate the mechanisms regulating the expression of heat-shock genes. In this review, we concentrate on this problem by analyzing the regulation in *E. coli,* because intensive studies during the past few years have led to the accumulation of data that may provide important clues for future studies. We place special emphasis on recent progress in understanding some of the initial events of the heat-shock response. For references to earlier work on the bacterial heat-shock response and on individual hsps, previous reviews dealing with more general or comprehensive subjects (27, 28, 31, 69, 71) should be consulted. Functional aspects of hsps, one of the most widely discussed topics in biology today, are treated in a number of recent reviews (17, 27, 107, 117).

INDUCTION OF HSPS AND THE HEAT-SHOCK REGULON

In *E. coli,* as in most other organisms, transient induction of hsps is a most conspicuous response of cells when they are exposed to a modest heat shock such as a shift from 30 to 42°C: most other proteins do not change their rates of synthesis appreciably, whereas still others decrease their synthesis transiently (50, 71, 112). The induction of hsps occurs almost immediately after temperature shift, reaches its maximum at about 5 min, and declines to a new steady-state level in 20–30 min. Induction also occurs coordinately at the level of transcription (110, 111). When cells are exposed to a lethal temperature such as 50°C, most proteins cease to be synthesized and those that remain are mostly hsps (12, 71).

A breakthrough in the discovery of heat-shock regulation in *E. coli* occurred in 1975, when Cooper & Ruettinger (10) reported a nonsense mutation affecting the synthesis of a major protein essential for growth at high temperature. This mutant, Tsn-K165, carried a mutation, *rpoH* (=*htpR, hin*), that was later shown by two laboratories to affect the synthesis of several proteins now known as hsps (68, 112). Most importantly, the *rpoH* gene product was identified as a class of the σ factors (σ^{32}) that are required for transcription of the heat-shock genes (32). These findings formed a basis for subsequent analyses of regulation and function of the heat-shock response in bacteria. At least some hsps can be induced by other agents, including DNA-damaging agents (45), amino acid analogues (30), stringent response (34), starvation for carbon source (37), λ phage infection (16, 43), unfolded proteins (30, 75), alkaline shift (90), and ethanol (98), suggesting that hsps may protect cells against a variety of environmental stresses.

The heat-shock regulon under σ^{32} control includes genes for about 20 proteins as defined by two-dimensional gel electrophoresis (69, 71). Most of the major *E. coli* hsps have homologous eukaryotic counterparts, e.g. DnaK, DnaJ, GrpE, GroEL, GroES, Lon, ClpP, ClpB, and HtpG (27, 107). Many of them (DnaK, DnaJ, GrpE, GroEL, GroES, and presumably HtpG) are chaperones that assist correct folding and assembly of proteins, and consequently affect diverse cellular processes including DNA replication (see 27), RNA transcription, protein transport (e.g. 6), proteolysis (e.g. 41, 87), flagella synthesis (80), and UV mutagenesis (15). These proteins, particularly when overproduced, can protect various other proteins from heat inactivation (e.g., 24, 61, 82), consistent with the genetic evidence for key roles of GroE and DnaK proteins (among other hsps) in supporting growth at physiological temperature (46, 49, 102). DnaK, DnaJ, and GrpE proteins, known to work synergistically in λ phage and P1 plasmid DNA replication (27), are also involved in many cellular functions including feedback control of the heat-shock response (31, 89). Besides their protective roles against protein inactivation in vivo, they reactivate denatured proteins such as RNA polymerase (82), as shown by experiments using purified proteins in vitro. On the other hand, at least three hsps represent either an ATP-dependent protease (Lon, La) (29) or a catalytic (ClpP, Ti) or regulatory (ATPase) subunit (ClpB) of another protease Clp (42, 44, 59, 84). These results are thus consistent with the notion that major functions of hsps are to prevent inactivation of cellular proteins, to reactivate once inactivated proteins, and to help degrade nonreparable denatured proteins that accumulate under normal growth as well as under stress conditions (27, 49, 107).

ROLE OF σ^{32} IN TRANSCRIPTION OF HEAT-SHOCK GENES

Induction of hsps upon temperature upshift occurs through the activation of transcription from promoters specifically recognized by RNA polymerase (E) that contains σ^{32} ($E\sigma^{32}$) (11, 32, 118). Genetic experiments with an *rpoH* mutant provided first evidence that the heat induction is positively regulated by the *rpoH* gene product. Mutations in *rpoH* prevented the transient increase in hsp synthesis following a temperature upshift without appreciably affecting synthesis of other proteins (68, 112). Moreover, the extent of induction of hsps depended on the amount of *rpoH* gene product synthesized (112). The finding that an amber mutant of *rpoD* (encoding σ^{70}) producing reduced amounts (10% of wild-type) of σ^{70} exhibited markedly increased synthesis of hsps (73) suggested the possibility that the *rpoH* gene product and σ^{70} compete

Table 1 Promoters transcribed by RNA polymerase containing σ^{32}

Map Position	Promoter	−35 Region	Spacing	−10 Region	Reference
0.3'	*dnaKp1*	T C T C c C C C T T G A t	14	C C C C A T t T A	11
0.3'	*dnaKp2*	T t g g g C a g T T G A A	13	C C C C t a t T A	11
10'	*lon*	T C T C g g C g T T G A A	14	C C C C A T a T A	11
10'	*clpP*	T g T t a t g C T T G A A	14	a C C C A T a a c	59[a]
11'	*htpGp1*	g C T C t C g C T T G A A	15	C C C C A T c T c	11
28'	*topAp1*	a C a a g g g g T T G A t	16	g t C C A T a T c	52
57'	*grpE*	T g c t t C C C T T G A A	14	C C C C A T a a t	56
57'	*clpB*	c a a t a a C C T T G A A	14	C C t C A T t T A	42
67'	*rpoDphs*	T g c C a C C C T T G A A	15	g a C g A T a T A	11
83'	*ibp*	g a T a a g g C T T G A A	13	a C C C A T t t t	2[a]
90'	*rrnBp1*	a t T t c C t C T T G t c	14	C C C t A T a a t	72[a]
94'	*groE*	T t T C c C C C T T G A A	13	C C C C A T t T c	11
F	*repE*	c C T g c t t a T T G A c	17	g a C a A T c T A	103
σ^{32} consensus		T C T C – C C C T T G A A	13 ~ 17	C C C C A T – T A	
σ^{70} consensus		T T G A C A	16 ~ 18	T A T A A T	

[a] To our knowledge, direct tests using an in vitro transcription system have not been reported for these promoters.

with each other for RNA polymerase core enzyme (115). Cloning and sequencing of the *rpoH* gene revealed that regions of its predicted gene product are strikingly similar to those of σ^{70} (47, 70, 97, 114). Grossman et al (32) purified the factor that permitted transcription of the heat-shock gene as a 32-kDa σ factor (σ^{32}) that specifically recognized the heat-inducible promoters located upstream of the heat-shock genes (5, 11, 23, 93).

The distinct promoter specificity of σ^{32} provides the basis for the characteristic heat inducibility of the regulon. At least some heat-shock genes (e.g. *rpoD, groE*), however, carry an additional promoter transcribed by $E\sigma^{70}$ (93, 118). At present, 13 promoters are known to be transcribed by $E\sigma^{32}$. Table 1 lists the map positions and sequences of these promoters. In general, these promoters are specifically recognized by $E\sigma^{32}$ but not by $E\sigma^{70}$ in vitro. Such a distinct specificity was also demonstrated in vivo, at least for *dnaKp1, dnaKp2, groE, htpGp1,* and *htpGp2* promoters, through the use of *rpoH* null mutants lacking σ^{32} (118): apparently neither σ^{70} nor other σ factors endow the *E. coli* RNA polymerase core enzyme with the ability to recognize these promoters.

Recent examination of the *rrnBp1* promoter revealed an unusual case: both -35 and -10 bp sequences of a σ^{32} promoter overlap those of the previously identified σ^{70} promoter, resulting in transcription from identical start sites by $E\sigma^{32}$ and $E\sigma^{70}$ (72a). Apparently similar paradoxical cases had been found in *Bacillus subtilis* where one promoter can be utilized by two forms of RNA polymerase (38, 91, 104). A consensus sequence derived from the present list of σ^{32} promoters, which agrees well with that reported previously (11), should be compared and contrasted with consensus σ^{70} promoters (Table 1). The list also includes the *repE* promoter of F plasmid encoding a replication initiator protein; the role of σ^{32} is not restricted to the chromosomal genes. This result exemplifies the critical involvement of σ^{32} in the maintenance of conjugative plasmids such as F (103).

ROLE OF σ^{32} IN CELL GROWTH

Although the essential role of σ^{32} in cell growth at high temperature was established when an *rpoH* mutant was first isolated (10, 68, 112), its role at low temperature was assessed in later experiments; the majority of temperature-sensitive *rpoH* mutants isolated in a strain lacking known tRNA suppressors turned out to result from nonsense mutations, suggesting that σ^{32} may be dispensable at relatively low temperatures (up to 34°C) (97, 114). Indeed, *E. coli* carrying the *rpoH* null mutations (caused either by a deletion or a transposon insertion) were isolated, though they failed to grow at temperatures above 20°C (118).

The role of σ^{32} in cell growth was further assessed through the isolation and characterization of temperature-resistant revertants of the Δ*rpoH* strain

(46). This analysis found that the inability of *rpoH* null mutants to grow at high temperature is primarily caused by the lack of sufficient amounts of key heat-shock proteins, GroEL and GroES. The reason for this is that hyperproduction of GroE (but not DnaK), either by the insertion of an exogenous promoter (an IS*10*-like element) into the *groE* upstream region or by the introduction of *groE*-expressing plasmids, enabled the mutant bacteria to grow at temperatures up to 40°C (46). Moreover, the amount of GroE proteins produced among the revertants or other derivatives of the Δ*rpoH* strain tested correlated well with the ability to grow at higher temperatures. These results implied that the residual amount of GroE produced in *rpoH* null mutants, owing to transcription from a minor σ^{70} promoter, supported their growth up to 20°C (118). Furthermore, when large amounts of both GroE and DnaK were produced either by promoter insertion(s) or by plasmid(s), the Δ*rpoH* strain could grow at up to 42°C (46).

Thus, at least in the absence of σ^{32}, GroE seemed to be essential at all temperatures, whereas DnaK appeared essential under more restricted conditions. Consistent with this is the finding that deletion mutants can be isolated for *dnaK* (7, 74) but not for *groE* (27). More recently, experiments designed to assess the effects of reduced GroE contents on protein metabolism were carried out under steady-state growth conditions (40). It was found that when the amount of GroE was specifically reduced to subnormal levels, the upper limits of growth temperature dropped significantly; the amounts of other hsps produced were elevated under these conditions. The different effects of GroE and DnaK observed with the Δ*rpoH* mutant (46) may reflect different functions of the two most abundant bacterial hsps (35), as indicated by many lines of evidence (e.g. 27, 49, 107), including the recent model of sequential function of DnaK (DnaJ, GrpE) and GroEL (GroES) in assisting the folding of nascent polypeptides in *E. coli* (48). Besides their inability to grow at high temperature, the *rpoH* mutants have been shown to be defective in proteolysis (4, 29), cell division (99), growth of temperate phages (T. Yura, unpublished result), and plasmid DNA replication (103). These phenotypes should result, at least in part, from reduced levels of one or more hsps.

REGULATORY ROLES OF σ^{32} IN THE HEAT-SHOCK RESPONSE

The mechanism underlying the rapid and transient increase in the rate of transcription of hsp genes in *E. coli* was intensively studied recently (31). In addition to its catalytic role in initiating transcription from heat-shock promoters, σ^{32} is directly involved in control at three different levels: synthesis, degradation, and activity. Thus, the levels and activity of σ^{32} play an active and dynamic role in regulating and fine tuning heat-shock gene expression under a variety of conditions (31, 33, 86).

Regulation of the σ^{32} Level

Investigators initially believed that σ^{32} may be activated at high temperature, because the heat induction was so rapid (111) as to preclude any models that assume increased synthesis of σ^{32} (e.g. 32). On the other hand, the extent of heat-shock induction of hsps depended on the amounts of *rpoH* gene product produced (112). Consistent with this, increasing the rate of σ^{32} synthesis without a corresponding temperature upshift also resulted in elevated synthesis of hsps (33). In addition, σ^{32} synthesized in vitro using a purified transcription-translation system directly activated transcription of heat-shock genes (5). Immunological detection of σ^{32} provided direct evidence that the cellular

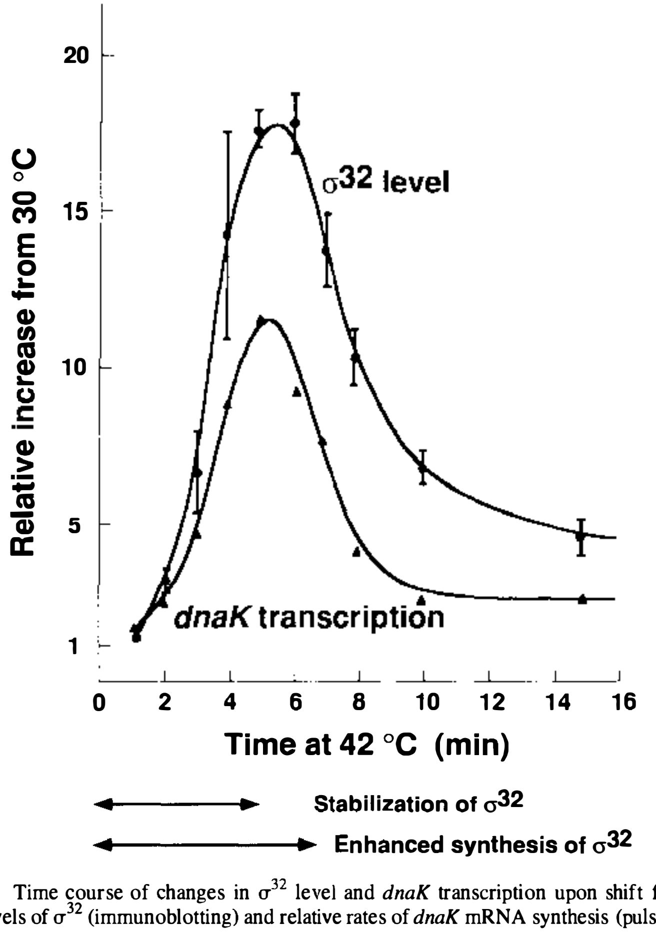

Figure 1 Time course of changes in σ^{32} level and *dnaK* transcription upon shift from 30 to 42°C. Levels of σ^{32} (immunoblotting) and relative rates of *dnaK* mRNA synthesis (pulse labeling) were determined. Reproduced with permission from Ref. 86.

levels of σ^{32} rapidly increase following a temperature upshift in both haploid cells (53, 86) and cells harboring an *rpoH*$^{+}$ multicopy plasmid (22). The ability of S-30 extracts to synthesize GroEL and DnaK proteins in vitro clearly reflected the intracellular σ^{32} levels in vivo at the time of cell harvesting: a much higher activity was obtained with cells that had been heat shocked (81).

Detailed time-course experiments using specific σ^{32} antisera were then conducted. When cells were shifted from 30 to 42°C, the intracellular concentration of σ^{32} increased 15- to 20-fold within 5 min and then declined to a new steady-state level severalfold higher than the preshift level (53, 86). The transient increase in σ^{32} levels during heat shock resulted from changes in both synthesis and stability of σ^{32} (86). Both the time course and the increase in σ^{32} levels were apparently sufficient to account for the increased transcription of heat-shock genes (Figure 1). The cellular concentration of σ^{32} during steady-state growth is very low, estimated to be only 10 to 30 molecules per cell (12). This is because σ^{32} is an extremely unstable protein (half-life = ~1 min) (33, 86, 96), and its synthesis is largely repressed at the level of translation (33, 39, 64, 86). Both the unusual instability and translational repression provide effective means of maintaining σ^{32} concentration at a minimum during steady-state growth when only low levels of hsps are needed. Following physiological or environmental stress, the levels of σ^{32} and of hsps are rapidly increased to meet the specific requirements of the given condition (33, 86).

Regulation of σ^{32} Stability

Perhaps one of the earliest events that occurs in response to heat-shock stress is the transient stabilization of otherwise unstable σ^{32}. As mentioned above, the half-life of σ^{32} produced from a single-copy *rpoH* gene during steady-state growth is about 1 min at both 30 and 42°C (86). During the initial studies, some variability in stability was found when σ^{32} was produced from various multicopy plasmids (3, 33, 96), presumably reflecting differences in strains, culture conditions, or other methods used. Apparently, σ^{32} is more stable when produced in larger amounts particularly at low temperatures such as 22°C (96).

Following a temperature shift from 30° to 42°C, σ^{32} is markedly though transiently stabilized for the first 4–5 min, which allows its rapid accumulation. After this initial phase, however, the marked instability of σ^{32} resumes, consistent with the transient increase in σ^{32} levels observed (Figure 1). The rapid stabilization of σ^{32} probably explains an almost instantaneous increase in the rates of hsp synthesis observed after a temperature upshift (111). Although little is known about the exact mechanisms underlying the extreme instability of σ^{32}, mutations in several genes as well as growth conditions have been shown to stabilize σ^{32}.

DnaK was first reported as a negative modulator of the heat-shock response, because a mutation in *dnaK* resulted in both elevated levels of hsps at low temperature (30°C) and prolonged synthesis of hsps upon shift to high temperature (42°C) (95). Other missense and null mutations of *dnaK* and those in two other hsp genes, *dnaJ* and *grpE,* were subsequently shown to exhibit similar phenotypes (89, 96). In all these mutants, σ^{32} was markedly stabilized, the half-life lasting 10- to 30-fold longer than that in wild-type bacteria. The increased stability but not the increased synthesis of σ^{32} accounts for the higher levels (severalfold) of hsps observed in these mutant backgrounds (89). Reduced cellular levels of GroE during steady-state growth also stabilized σ^{32} significantly and resulted in elevated synthesis of other hsps, though the effect was less striking than those observed with the *dnaK, dnaJ,* or *grpE* mutants (40). Because none of these hsps are known to be proteases, they most likely assist proteolysis directly by binding to σ^{32}, as recently shown for DnaK, DnaJ, and GrpE (25, 54), and presenting it to an unidentified protein-degradation system. Alternatively, the hsps may work indirectly through their potential effect on protease activation. *E. coli* cells infected with phage λ transiently produce high levels of hsps (16, 43) because of the stabilization of σ^{32}; the λ*c*III gene is apparently responsible for this stabilization, because overproduction of cIII protein alone enhanced hsp synthesis in uninfected bacteria (3). Recent work on the expression and stability of various σ^{32}-β-galactosidase fusion proteins using a *rpoH-lacZ* gene suggested that a specific segment of σ^{32} is required for regulation of σ^{32} degradation and feedback control of the heat-shock response (see below).

Regulation of σ^{32} *Activity*

Although an increase in σ^{32} level primarily causes the induction of hsps upon temperature upshift, the rate of hsp synthesis does not always reflect the intracellular levels of σ^{32} and can be reduced by inhibiting σ^{32} activity. Such a situation arises when excess hsps are synthesized following an initial overproduction of σ^{32}: whereas σ^{32} continues to be made at high levels, enhanced synthesis of hsps is observed only transiently (33). Also, the level of σ^{32} in the *dnaK, dnaJ,* or *grpE* mutants is 10- to 20-fold higher than in the wild-type, but the synthesis of hsps is only 2- to 4-fold greater, indicating that the activity of σ^{32} is inhibited (89). In addition, when *E. coli* cells grown at high temperature (42°C) are shifted to low temperature (30°C), σ^{32} levels do not appreciably decrease. However, both transcription of heat-shock genes and synthesis of hsps decrease markedly, though transiently, indicating that the activity of σ^{32} is transiently inhibited (88, 92). Furthermore, the same subset of hsps (DnaK, DnaJ, and GrpE) involved in the control of σ^{32} stability is apparently involved in the control of σ^{32} activity (C. Gross, personal communication; N. Kusukawa & T. Yura, unpublished result).

Recent experiments from two laboratories revealed that σ^{32} physically binds to DnaK, DnaJ, or GrpE. Using purified proteins, researchers showed that σ^{32} binds to wild-type DnaK, but not to DnaK756 mutant protein in vitro and that the resulting complex is disrupted by Mg^{2+}-ATP (54). Complexes between these hsps and σ^{32} are also found in vivo: histidine-tagged σ^{32} was shown to be associated with each of these hsps, apparently independently, and the complexes between σ^{32} and DnaK (or GrpE) or between σ^{32} and DnaJ behaved differently in the presence or absence of ATP (25). In further in vitro binding studies, a stable ternary complex DnaJ-σ^{32}-DnaK was formed effectively only in the presence of GrpE or ATP. Moreover, the set of three hsps was found to interfere with the ability of σ^{32} to bind to core RNA polymerase, and hence its ability to initiate transcription from heat-shock promoters (C. Georgopoulos, personal communication). These results taken together indicate that DnaK, DnaJ, and GrpE proteins work synergistically to interfere with the activity of σ^{32}. Such an apparent sequestration of σ^{32} from core RNA polymerase could presumably render σ^{32} a better substrate for one or more proteases, thus linking the control of activity to that of σ^{32} stability (27).

REGULATION OF σ^{32} SYNTHESIS

Transcription of the rpoH *Gene*

The *rpoH* gene has at least four promoters; three (P1, P4, P5) are transcribed by Eσ^{70} (19, 21, 63), whereas one (P3) is transcribed by RNA polymerase containing a novel σ factor, σ^E (=σ^{24}), that is especially active at very high temperature (18, 106). Figure 2 shows the organization of *rpoH* promoters. The most distal P1 promoter is the strongest, and together with P4, is responsible for most *rpoH* transcription under most growth conditions. The activity of P1 does not seem to be highly regulated, but the fact that its −35 region is located within the adjacent *ftsX* gene suggests a possible coupling with *fts* gene expression, perhaps mediated by the DnaA protein (13). In contrast, the activity of P4 is enhanced 2- to 3-fold upon shift from low to

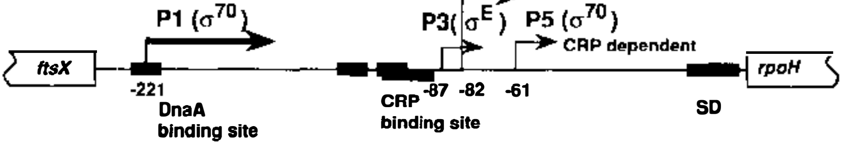

Figure 2 Organization of *rpoH* promoters and putative regulatory sites (taken from 19, 21, 63). The numbering system of promoters is that used in Refs. 19 and 63.

high temperature (19, 21). A pair of DnaA protein–binding consensus sequences found upstream of the P3 and P4 promoters were indeed shown to bind DnaA protein and specifically inhibit transcription from these promoters in vivo and in vitro (105). P5 is a weak promoter, enhanced in the absence of glucose or by addition of ethanol, and requires cAMP and cAMP-receptor protein for its function in vitro (63). However, P5 is unlikely to be involved in the *rpoH*-dependent induction of hsps during carbon starvation, because the induction is not cAMP-CRP (cAMP receptor protein) dependent (78).

The P3 promoter seems to be particularly important, because it becomes highly active under unusual conditions such as exposure to a lethal temperature (50°C) at which σ^{70} (and perhaps other σ factors) is inactivated. This result probably explains the observation that only hsps are highly and continuously expressed at 50°C (12, 71). Several other bacterial promoters carry a sequence similar to *rpoHp3,* and one of them, *htrA,* is actually transcribed in vitro by $E\sigma^{E}$ (18, 57). The *htrA* (*degP*) gene encodes a periplasmic protease that is important for survival at high temperature (55, 85). These results suggest that genes whose transcription depends on σ^{E} constitute a second heat-shock regulon of *E. coli* and that their function may strengthen or complement that of the σ^{32} regulon in coping with heat shock or other stresses (31). For example, products of these genes may be required for thermotolerance; cells that had been pretreated at 42°C acquire tolerance to a subsequent exposure to lethal temperature (55°C) (112), whereas overproduction of σ^{32} (and hsps) alone without a prior temperature upshift will not result in thermotolerance (100).

In spite of such a complex organization and potential regulatory importance of the various promoter regions in *rpoH* transcription, the bulk of the transient increase in σ^{32} synthesis observed upon temperature upshift (30° to 42°C) results from an increase in *rpoH* translation, rather than transcription. The major function of transcriptional control therefore appears to be to provide appropriate levels of *rpoH* mRNA under a variety of steady-state growth conditions, as well as to maintain the critical levels of *rpoH* message so as to tolerate some extremely stressful conditions such as lethal temperature and harmful agents, including ethanol.

Translational Induction of σ^{32} Synthesis

Early observations of increased levels of *rpoH* mRNA upon exposure to high temperature and of increased transcription from some *rpoH* promoters at high temperature (19, 21, 94) suggested that most of the heat-induced synthesis of σ^{32} might occur at the level of transcription. However, other results favored translational control, and the elevated levels of *rpoH* mRNA that accumulate after temperature upshift mostly result from increased mRNA stability caused by increased translation, rather than increased transcription. The following evidence indicates that the increased synthesis of σ^{32} observed after shift from 30 to 42°C occurs primarily at the translational level: First, the synthesis of σ^{32}, but not of *rpoH* mRNA, increases markedly after a temperature upshift (19). Second, heat-induced synthesis of σ^{32}-β-galactosidase fusion protein depends on the particular translational initiation region of *rpoH* used, not on the promoter region (64, 86). Third, the increase of *rpoH* mRNA level is preceded by the increase of σ^{32} synthesis: the peak rate of σ^{32} (or fusion protein) synthesis and the maximal accumulation of *rpoH* mRNA are seen at 3 and 6 min, respectively, after a temprature shift (39, 86). Finally, heat-shock induction of σ^{32}-β-galactosidase fusion protein occurs even when RNA synthesis is inhibited by rifampicin (64).

CIS-ACTING REGULATORY ELEMENTS ON *rpoH* mRNA Attempts to localize the *rpoH* region(s) that are involved in the thermoregulation of σ^{32} synthesis (39, 64) followed the studies mentioned above. Extensive analysis of 3′- and 5′-deletion derivatives of an *rpoH-lacZ* gene fusion (GF364 containing 364 nucleotides of *rpoH* fused to *lacZ*) led to the identification of two *cis*-acting

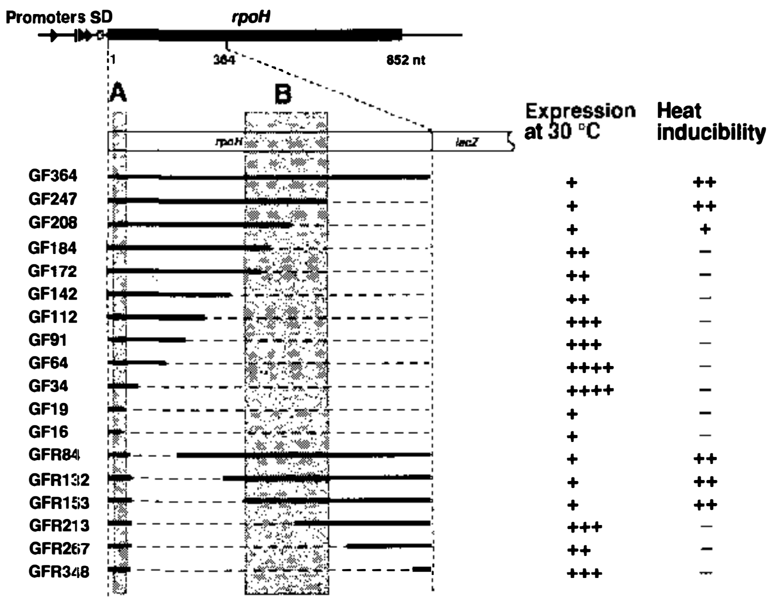

Figure 3 Structure and expression of deletion derivatives of *rpoH-lacZ* gene fusion (GF364). Cells (Δ*lac*) lysogenic for λ carrying a 3′- (GF) or 5′-deletion (GFR) derivative of GF364 were examined for expression at 30°C and at 42°C, with immunoprecipitation of pulse-labeled fusion proteins. Regions A and B represent *cis*-acting elements involved in heat induction of *rpoH* translation. Reproduced with permission from Ref. 65.

mRNA regions within the *rpoH* coding sequence. One region, designated A, corresponds to nucleotides 6–20, found immediately downstream of the initiation codon, and acts as a positive element enhancing the rate of translation: deletions extending into this region from the 3′ end exhibited drastically reduced expression (~15-fold) (Figure 3). This sequence is similar to the downstream box (83), which enhances expression of gene 0.3 of phage T7 and is complementary to nucleotides 1469–1483 of 16S rRNA. The downstream box, now found in several bacterial genes (20, 79, 83), presumably stimulates translation by stabilizing the binding of message to the 30S ribosomal subunit (79). Consistent with this expectation, base substitutions that either reduced or increased complementarity to 16S rRNA, decreased or increased the expression, respectively, in all cases tested (20, 64, 79).

The other internal *rpoH* region, designated B, is located between nucleotides ~153 and ~247 and acts as a negative element in thermoregulation, because deletions of this region from the 5′ or 3′ end resulted in partially or fully constitutive expression, i.e. increased expression at low temperature (30°C); the extent of induction upon shift to 42°C was concomitantly reduced as well (39, 64) (see Figure 3). Interestingly, some base-substitution mutations within region A exhibited constitutive expression (64, 116), indicating that besides controlling the basal level of σ^{32} expression, region A may also affect thermoregulation. Moreover, frameshift mutations that drastically alter the amino acid sequence of region B but hardly change its nucleotide sequence did not substantially interfere with gene regulation (64). These results taken together provide strong evidence that the 5′ portion of *rpoH* mRNA that corresponds to some 200 nucleotides of coding sequence is critically involved in controlling thermal induction of σ^{32} synthesis.

THE mRNA SECONDARY STRUCTURE MODEL The above information leads one to ask how translation of *rpoH* mRNA is actually regulated. Potential secondary structures of *rpoH* mRNA were examined using a computer program (36), and a plausible structure, having minimal free energy, emerged among the various alternative folded structures (Figure 4). According to this model, part of region A plus the initiation codon and part of region B can base pair, leading to a sequestration of the initiation codon. Such a secondary structure may be transiently destabilized or disrupted when cells are exposed to high temperature, thus permitting higher rates of translation initiation. Consistent with such a proposal, all deletion derivatives of an *rpoH-lacZ* gene fusion that exhibited constitutive expression were predicted to result in a loss of base pairings around the initiation region. Furthermore, random base substitution–type mutations selected on the basis of constitutive expression of an *rpoH-lacZ* gene fusion were shown to be localized within region A and a

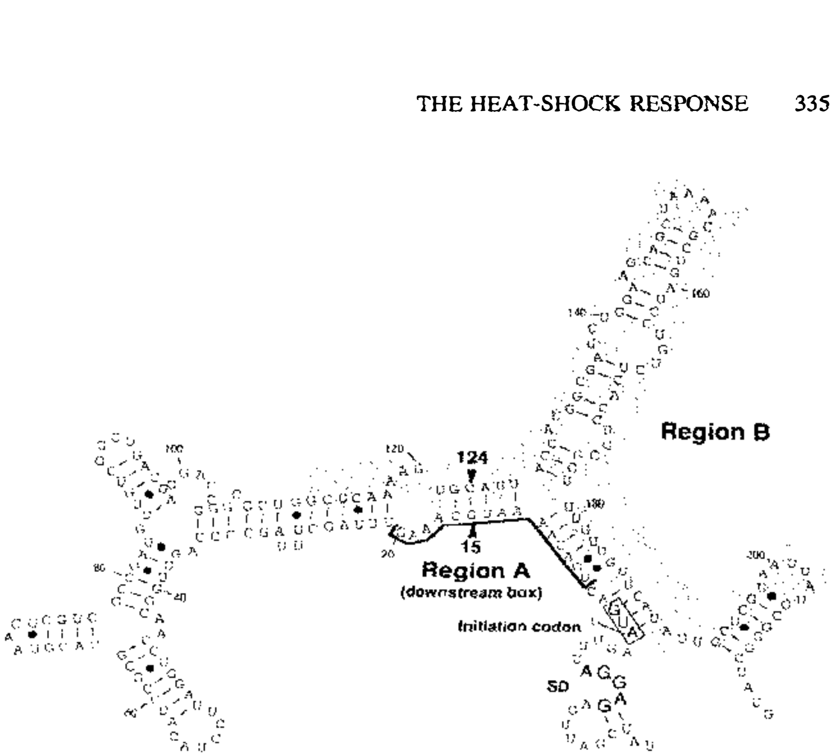

Figure 4 Predicted secondary structure of the 5′ portion of *rpoH* mRNA. The region B shown here (*shaded*) is displaced ~40 bp from that originally defined by the deletion analysis shown in Figure 3 (see text) (adapted from 64).

region spanning positions 112–208 (a shaded area in Figure 4, designated region B); the latter region, however, is shifted toward the 5′ end by ~40 nucleotides, as compared with region B originally defined by the deletion analysis (116) (cf Figures 3, 4).

Examination of additional base substitution mutations that are predicted to disrupt the secondary structure, and the putative compensatory mutations that are predicted to restore the secondary structure, further substantiated the above model (116). For example, base pairings between nucleotides 6U and 184G, 15G and 124C, 16C and 123G, and 17A and 122U appeared to be important, because single base changes predicted to disrupt a base pairing at each of these positions resulted in virtually constitutive expression. At least in some instances, compensatory changes predicted to restore the base pairing indeed resulted in an essentially parental phenotype (6G-184C; 15A-124U). Moreover, other combinations of bases at positions 15 and 124 that do not permit base pairing failed to restore the parental phenotype. The results obtained with the 15-124 pair of bases shown in Table 2a illustrate the point.

Table 2 Effects of compensatory and noncompensatory base substitution mutations on fusion protein synthesis[a]

Strain	mRNA base change	Fusion protein synthesis: Synthesis rate (30°C)	Fusion protein synthesis: Induction at 42° (x-fold)
a			
Wild-type	—	1.0[a]	5.5
15A	G → A	6.2	1.5
124U	C → U	3.8	2.3
15A / 124U	G–C → A–U	1.5	5.3
124G	C → G	3.6	1.1
15A / 124G	G–C → A–G	4.2	1.2
124A	C → A	2.3	1.5
15A / 124A	G–C → A–A	4.6	1.1
b			
15C	G → C	3.1	2.2
124G	C → G	2.7	2.0
15C / 124G	G–C → C–G	1.2	1.3

[a] See legend to Figure 3 for the methods used for determination of fusion-protein synthesis.

Interestingly, these analyses revealed other instances in which effects of single-base changes that cause constitutive expression were compensated for with respect to repression at low temperature but not compensated for with respect to heat inducibility (15C–124G; 17C–122G). The latter observation (Table 2b) suggests a possible involvement of a transacting (protein) factor that could modulate the stability of mRNA secondary structure by binding to the sequence including nucleotides 15–17, and/or 122–124. Such a hypothetical factor may be either a positive factor (activator) that would destabilize mRNA secondary structure upon heat shock, or a negative factor (repressor) that would stabilize the secondary structure under nonstress conditions and be inactivated upon heat shock.

A few additional internal deletions have been examined (116). A deletion of 12 nucleotides (from 178 to 189), predicted to disrupt a major stem structure involving the translation-initiation region, resulted in fully constitutive expression as expected from the present model. Another deletion of 51 nucleotides (127–177), predicted to delete one of the major stem structures (Figure 4) but to retain the rest of the secondary structures, exhibited significantly reduced expression at 30°C and failed to be induced at 42°C, suggesting that the deletion resulted in an overstabilization of the remaining portions of the mRNA structure (a superrepressed phenotype). These results, together with those of base substitution analysis, are consistent with the notion that thermoregulation of σ^{32} synthesis is mediated by a tight and possibly subtle adjustment of *rpoH* mRNA secondary structure, and lend further support to, though do not prove, a model such as the one depicted in Figure 4.

SEARCH FOR FACTOR(S) CONTROLLING *rpoH* TRANSLATION As to the possible transacting factor(s) involved in translational induction of σ^{32} synthesis, the heat-shock proteins DnaK, DnaJ, and GrpE that exert negative control on the heat-shock response (89, 95, 96) are not required for heat induction of σ^{32} or fusion protein synthesis: essentially normal expression at 30°C and heat induction at 42°C was observed in mutants lacking DnaK and/or DnaJ functions (64, 89). Also, autogenous regulation of either repression under steady-state growth or heat induction by σ^{32} itself is excluded, because deletion of *rpoH* had no appreciable effects on either of these processes (64). Indeed, the latter results make it unlikely that any of the heat-shock proteins under σ^{32} control are involved.

An extragenic suppressor of *rpoH,* called *suhB,* suppresses the temperature-sensitive growth of *rpoH15* missense mutant, endows the strain with a cold-sensitive growth phenotype, and enhances the synthesis of mutated σ^{32} at the translational level (113). The fact that *suhB* does not enhance σ^{32} synthesis in the *rpoH*$^{+}$ strain suggests that the wild-type SuhB protein represses *rpoH* mRNA translation when one or more hsps are under-supplied. The cold-sensitive growth of a *suhB* missense mutant, but not of a *suhB* null mutant, can be suppressed by an *rnc* mutation affecting RNase III (which specifically degrades dsRNA), suggesting an interaction between SuhB and RNase III (Y. Nakamura, personal communication). In agreement with the sequence similarity between SuhB and eukaryotic inositol monophosphate phosphatases (72), extracts of *E. coli* carrying a multicopy *suhB*$^{+}$ plasmid were recently found to contain elevated levels of phosphatase activity (K.

Shiba, personal communication). Although *suhB* could participate in modulating σ^{32} synthesis under certain conditions, the precise mechanism remains unknown.

Another temperature-sensitive mutant, *htrC,* whose product is essential for cell viability only at high temperature, shows constitutive synthesis of hsps and elevated levels of heat-shock gene transcripts (77), suggesting that it may modulate the translational control of σ^{32} synthesis. Contrary to such expectations, the *htrC* mutation must affect the stability of σ^{32}, because it exerted no effect on the rate of synthesis of the GF364 or GF807 *rpoH-lacZ* fusion proteins (H. Nagai, unpublished result).

Translational Repression of rpoH *mRNA: Feedback Control of* σ^{32} *Synthesis*

The first suggestion that the shut-off of σ^{32} synthesis, following its initial induction, occurs posttranscriptionally came from the observation that a repression of σ^{32} synthesis followed its overproduction by the λP_L promoter at 42°C, while its transcription continued at high levels for a long period (33). Subsequently, the shut-off of heat-induced synthesis of an *rpoH-lacZ* fusion protein (a fusion similar to GF807) was observed several minutes prior to the decrease in mRNA level (39). Thus, repression as well as the initial induction of σ^{32} synthesis seen after a temperature upshift occurs primarily at the level of translation, presumably reflecting the ability of the *rpoH* mRNAs to be translated.

Examination of the translational control of fusion-protein synthesis using *rpoH-lacZ* gene fusions revealed striking differences in the kinetics of expression between fusion GF807, which contained most of the *rpoH* coding sequence (807 out of 852 nucleotides), and GF364, which contained the N terminus–coding 364 nucleotides, used in most previous work. Whereas a marked induction followed by clear shut-off was observed with GF807, as with the synthesis of authentic σ^{32} (see Figure 1), similar induction but very little shut-off was observed with fusion GF364. This difference in σ^{32} synthesis is not merely a reflection of the differing stability of the fusion proteins produced (see below). When expression of fusion protein from GF807 was examined in the Δ*dnaK52* or *dnaJ259* mutant bacteria, the induction occurred normally but failed to shut-off (66), as previously found with authentic σ^{32} (89). The appreciable shut-off observed with GF364 in some earlier experiments (64) is probably explained by the assumption that the cultures used in these studies had already entered the stationary phase.

Examination of a set of in-frame 3′-deletion derivatives of GF807 led to the further localization of the *rpoH* region specifically involved in the shut-off of σ^{32} synthesis (66). Deletions retaining 433 or more nucleotides of the 5′

Table 3 Expression and stability of fusion protein with 3' deletion derivatives of gene fusion GF807[a]

Gene fusion numbers	5' *rpoH* sequence contained in fusion	Heat induction	Shut off	Stability at 30°C (half-life, min)
GF807	807	+	+	~1
GF712	712	+	+	~1
GF661	661	+	+	~1
GF533	533	+	+	~2
GF439	439	+	+	~2
GF433	433	+	+	~2
GF364	364	+	−	~20
GF247	247	+	−	~20

[a] See legend to Figure 3 for the methods used.

rpoH sequence shut off normally, whereas those carrying 364 or fewer 5′ nucleotides failed to shut off (Table 3). Thus, the small segment flanked by nucleotides 364 and 433 (coding for amino acid residues 122–144) appeared to be important for the normal shut-off of σ^{32} synthesis. Furthermore, a frameshift mutation of fusion GF807 carrying an amino acid sequence drastically altered specifically for this 23–amino acid segment showed little shut-off following initial induction. Evidently, a distinct segment of the σ^{32} polypeptide, designated C, is critically involved in the DnaK (DnaJ, GrpE)-mediated shut-off of fusion-protein synthesis, and presumably of σ^{32} synthesis, although involvement of other segment(s), particularly those located 5′ to segment C, cannot be excluded at present (66).

Possible Coupling Between Translational Repression and Degradation of σ^{32}

As previously noted, the fusion protein produced by GF807 is as unstable (half-life 1 min) as authentic σ^{32}, whereas the protein made by GF364 is quite stable (half-life 20 min) (64, 66), suggesting that a region(s) located in the C-terminal half of σ^{32} is important for its characteristic instability. A striking correlation was found between stability and shut-off of fusion proteins when the set of 3′-deletion derivatives of GF807 was examined: the same segment of σ^{32} critical for translational repression appeared to be important for its instability as well (Table 3). This rather unexpected result suggested that the shut-off of σ^{32} synthesis following heat induction and the degradation of σ^{32} may share some common mechanism or factor(s).

Several interesting possibilities raised by the above finding are that: a distinct segment of σ^{32} is critically involved in both translational repression and degradation of fusion protein and presumably of σ^{32}; this segment might represent a site of interaction with the DnaK/DnaJ proteins; and translational repression might be triggered by an interaction with DnaK/DnaJ during polypeptide chain growth, which may in turn inhibit further translation of *rpoH* mRNA.

Although it is not known how the DnaK/DnaJ-mediated shut-off of σ^{32} synthesis is related to σ^{32} degradation, a direct physical interaction between DnaK/DnaJ and the nascent σ^{32} polypeptide chain (or fusion protein) may cause both shut-off of synthesis (translational attenuation?) and degradation of σ^{32} polypeptides, as judged by the fact that these hsps and σ^{32} physically interact in vitro (54). Such an interaction could eventually lead to a repression of translation and rapid degradation of σ^{32} by an as yet unknown mechanism(s). Whatever the actual mechanism may turn out to be, there appears to be a dynamic link between the feedback controls of synthesis and degradation of σ^{32} that would play important roles in regulation of the heat-shock response.

THE FEEDBACK REGULATORY CIRCUITS

The rapid accumulation of hsps upon temperature upshift brought about by increased σ^{32} levels is shortly followed by an adaptation period during which the levels of σ^{32} and of hsps are readjusted to the new steady-state growth condition at higher temperature. The readjustment includes a shut-off of heat-induced synthesis of σ^{32} and resumed degradation of σ^{32} within several minutes after temperature shift. The central theme in this regulatory circuit is a negative feedback control exerted by a set of heat-shock proteins (DnaK, DnaJ, and GrpE) that mediate normal shut-off of heat-induced synthesis and destabilization of σ^{32} (89, 95). The same set of hsps can inhibit the activity of σ^{32} by interfering with its association with core RNA polymerase in vitro (C. Georgopoulos, personal communication), suggesting that similar modes of hsp functions may operate in vivo. Thus, the feedback control appears to be exerted at three distinct steps, namely, synthesis, degradation, and activity of σ^{32}, all contributing to a precise and prompt readjustment of the intracellular concentrations of hsps that are optimal for a given set of physiological conditions.

Such apparently multiple feedback regulatory circuits could work independently from one another. However, as we discussed above, the control of synthesis and degradation of σ^{32} during the shut-off phase is probably mediated by the distinct segment of the σ^{32} polypeptide, suggesting that the two processes may be interconnected. Segment C includes a region, spanning regions 2.4 and 3.1, that is conserved relatively little among bacterial σ factors (Figure 5*a*) (58). It is also flanked by segment 2.1, which is thought to be mainly responsible for binding to core RNA polymerase (51) and the region

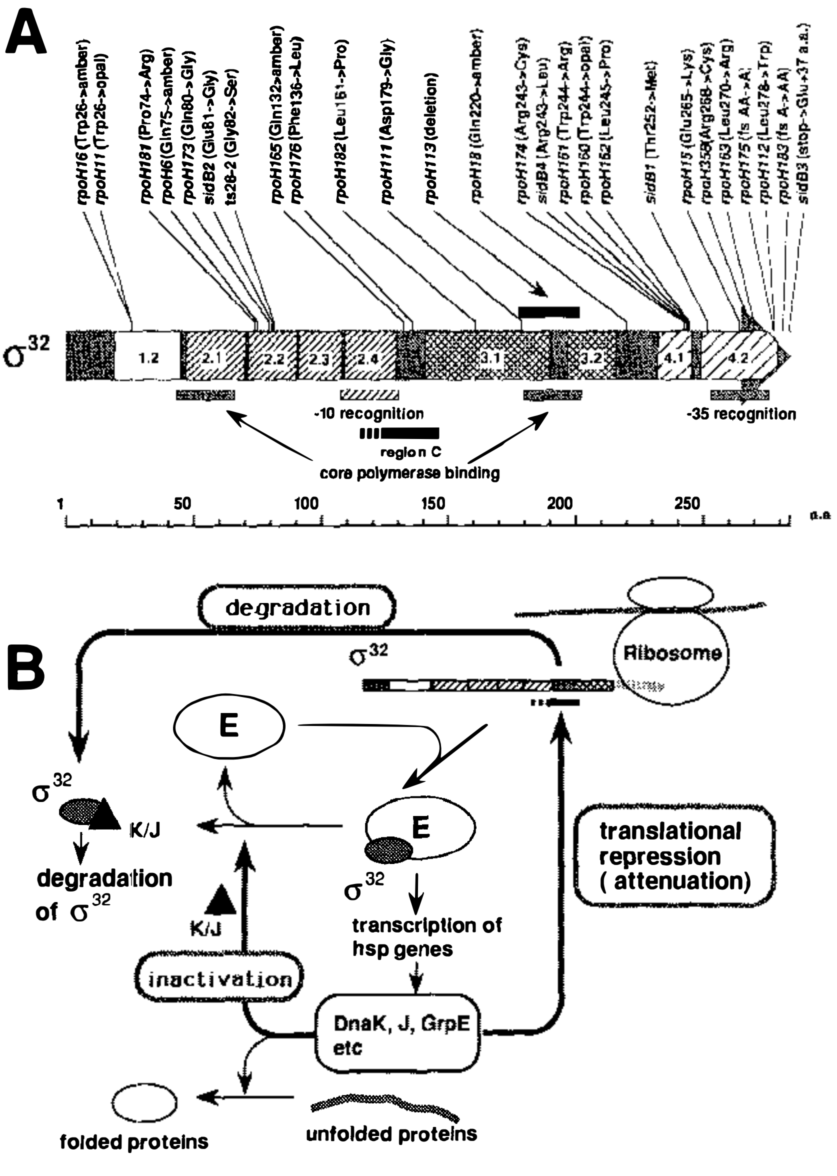

Figure 5 (*A*) Structural organization of σ^{32} protein indicating conserved regions [shown by numbers (see 58)] and mutations of known positions (taken from 8, 9, 72b, 114; R. Calendar, personal communication). (*B*) A schematic model that illustrates the DnaK (DnaJ, GrpE)-mediated feedback control circuits operating at three distinct but interconnected levels: synthesis, degradation, and activity of σ^{32}.

recently shown to play an auxiliary role in core polymerase binding (9, 119). Thus, when the N-terminal portion of nascent σ^{32} polypeptide is synthesized to somewhere beyond segment C on the ribosome, it may provide two separate but overlapping segments for binding to core RNA polymerase on one hand, and to DnaK/DnaJ proteins on the other: these bindings may be mutually exculsive. Accordingly, we can envisage the following series of events that might take place during the heat-shock response.

First, during steady-state growth at low temperature, σ^{32} synthesis is largely repressed at the level of translation initiation, and the σ^{32} produced is rendered unstable through binding to DnaK/DnaJ rather than to core RNA polymerase. Second, when cells are shifted to a higher temperature, a sudden decrease in free DnaK/DnaJ, owing to binding to unfolded/denatured proteins, will rapidly reduce their binding to σ^{32} and facilitates the association of σ^{32} with core RNA polymerase, thus stabilizing σ^{32}. In parallel with this stabilization, synthesis of σ^{32} is induced through a transient disruption of its mRNA secondary structure, and most of the nascent σ^{32} polypeptides synthesized will bind to the core RNA polymerase, rather than to DnaK/DnaJ, thus enhancing transcription from heat-shock promoters. Third, as the levels of free DnaK/DnaJ become sufficiently high, they will resume binding to σ^{32}, repress or attenuate translation, destabilize σ^{32}, and prevent the association of σ^{32} with core RNA polymerase; this would lead to the establishment of a new equilibrium appropriate for steady-state growth. Such regulatory circuits are schematically shown in Figure 5*b*.

CELLULAR SENSORS AND SIGNALS

Although the nature of sensors and signals in the heat-shock response is presently unknown, it has recently been the subject of much discussion and speculation. An earlier proposal suggested that increased amounts of abnormal proteins formed by heat-shock stress would titrate the function of protease Lon (La), an hsp required for degradation of abnormal proteins, leading to transient stabilization of σ^{32} (30). However, purified Lon protease does not seem to degrade σ^{32} (cited in 31), and instead DnaK, DnaJ, and GrpE proteins facilitate in vivo degradation of σ^{32}, probably by presenting it to unknown protease system(s). Because the concentrations of both hsps and proteolytic substrates probably do not increase appreciably upon mild heat shock, the extreme sensitivity of hsp function to changes in protein-substrate concentration could be explained by assuming that the hsps are involved primarily in polymeric interactions (31). The fact that DnaJ forms dimers (120), and recent genetic evidence suggesting that at least one form of DnaK is oligomeric (109), are consistent with such a possibility.

Another recent proposal is that the free pool of DnaK (and its eukaryotic homologue, HSP70) serves as a cellular thermometer that monitors changes in cellular concentration of unfolded or denatured proteins and regulates the expression of all hsps in prokaryotes and eukaryotes (12). Besides this indirect one, another role for DnaK as a direct thermometer was proposed on the basis of the extremely sharp temperature dependency of its autophosphorylation and ATPase activities (60). Either or both of these mechanisms may operate in sensing changes in the unfolded/denatured states of cellular proteins caused by heat and other stresses. Alternatively, DnaJ rather than DnaK was postulated to play a key regulatory function in the heat-shock response (25). The exact roles of DnaK/DnaJ and other hsps in transmitting heat-shock signals to σ^{32} at all steps of regulatory circuits would be a central issue for future studies. An entirely different model in which ribosomes are assumed to serve as the sensor for both heat shock and cold shock has been proposed (101).

A putative regulatory circuit for controlling translational induction of σ^{32} seems to offer a particular challenge because of its apparently independent nature from the rest of the regulatory circuits. Clearly, DnaK and other hsps are not directly involved in translational induction of σ^{32} observed immediately upon a temperature upshift, but instead the hypothetical regulator(s) of the *rpoH* mRNA secondary structure are likely to receive a signal generated by heat shock, rather than through titration of hsp function. Ethanol induces the synthesis of fusion protein from GF364, as well as the synthesis of authentic σ^{32} (86), suggesting that ethanol and high temperature generate similar signals in causing translational induction of σ^{32} synthesis (H. Yuzawa, unpublished result).

In contrast to the translational induction that occurs apparently independently of hsp levels, the shut-off of σ^{32} synthesis or translational repression and destabilization of σ^{32} provide major targets for negative feedback control that specifically requires the synergistic functions of DnaK, DnaJ, and GrpE. In light of what seems to be a critical involvement of a specific segment (amino acid sequence) of the σ^{32} polypeptide, direct physical interaction between one or more hsps and σ^{32} might occur during polypeptide chain elongation. In addition to, or instead of, such translational attenuation, however, direct interaction between intact σ^{32} (or σ^{32} bound to core RNA polymerase) and hsps, as suggested by experiments in vitro (C. Georgopoulos, personal communication), could also occur under certain conditions. Such an interaction would account for the inhibition of σ^{32} activity observed when σ^{32} (and hsps) is constantly overproduced by an exogenous promoter (33) or through a temperature downshift (88, 92). It might also play an auxiliary role during the shut-off phase of the heat-shock response: the slight delay in the shut-off kinetics of σ^{32} level as compared to that of *dnaK* transcription (see Figure 1) is actually in agreement with such an interpretation.

PERSPECTIVES

Intensive studies on the synthesis, stability, and activity of σ^{32} (or σ^{32}-β-galactosidase fusion protein) upon temperature upshift, coupled with a better understanding of the functional roles of hsps in recent years, have offered new insights into the mechanisms of regulation of the heat-shock response in *E. coli.* Further biochemical and genetic studies on the regulatory events occurring during the induction and shut-off phase should continue to provide basic and useful information for further understanding of early reactions related to sensors and signals of the heat-shock response. Both translational induction mediated by *rpoH* mRNA secondary structures and repression (or attenuation) modulated by DnaK and other hsps require further investigation. These studies will be important not only for elucidating the regulatory mechanism of the heat-shock response but also for further exploitation of the involvement of protein-coding sequence in translational control of gene expression in bacteria. The intriguing possibility that there are two separate pathways for the heat-shock signal transduction, one acting on translational induction of *rpoH* mRNA and the other acting on DnaK/DnaJ-mediated negative controls on translational attenuation and stabilization/destabilization of σ^{32}, should soon be investigated directly.

Among the other approaches of numerous laboratories to the regulation of the heat-shock response are biochemical studies on the interaction between σ^{32} and DnaK/DnaJ and genetic and biochemical analyses on the structure and function of σ^{32} and several important hsps such as DnaK and DnaJ. These lines of investigation should provide information essential for our ultimate understanding of the mechanisms underlying the various regulatory events along the signal-transduction pathways. However, additional factors that cannot yet be predicted from our current picture of the regulation of the heat-shock response will probably be uncovered in the near future.

Finally, in view of the recent advances in the field of heat-shock response in *E. coli,* some comparative studies with other bacteria are warranted. So far, only a few instances of *rpoH* or σ^{32} homologues have been reported in the literature (1, 26). Structural and functional analyses of heat-shock genes with *B. subtilis, Clostridium acetobutylicum,* and others led to the suggestion that the regulation of the heat-shock response is mediated by an inverted repeat consensus sequence found in front of these genes (e.g. 67, 108). It would be both interesting and rewarding to compare similarities and differences in the modes and mechanisms of heat-shock regulation within prokaryotes as well as between prokaryotes and eukaryotes.

ACKNOWLEDGMENTS

We are grateful to Carol A. Gross and her associates for invaluable advice and cooperation during past years that was instrumental to much of our recent work discussed here. We also thank C. Georgopoulos, K. Ito, T. Nagata, K. Shigesada, G. C. Walker, and other colleagues for kind advice and for sharing unpublished results with us for the writing of this review, and M. Mihara for help in preparing the manuscript.

Literature Cited

1. Allan, B., Linseman, M., McDonald, L. A., Lam, J. S., Kropinski, A. M. 1988. Heat shock response of *Pseudomonas aeruginosa*. *J. Bacteriol.* 170: 3668–74
2. Allen, S. P., Polazzi, J. O., Gierse, J. K., Easton, A. M. 1992. Two novel heat shock genes encoding proteins produced in response to heterologous protein expression in *Escherichia coli*. *J. Bacteriol.* 174:6938–47
3. Bahl, H., Echols, H., Straus, D. B., Court, D., Crowl, R., Georgopoulos, C. P. 1987. Induction of the heat shock response of *E. coli* through stabilization of σ^{32} by the phage λ cIII protein. *Genes Dev.* 1:57–64
4. Baker, T. A., Grossman, A. D., Gross, C A. 1984. A gene regulating the heat shock response in *Escherichia coli* also affects proteolysis. *Proc. Natl. Acad. Sci. USA* 81:6779–83
5. Bloom, M., Skelly, S., VanBogelen, R., Neidhardt, F., Brot, N., Weissbach, H. 1986. In vitro effect of the *Escherichia coli* heat shock regulatory protein on expression of heat shock genes. *J. Bacteriol.* 166:380–84
6. Bochkareva, E. S., Lisson, Girshovich, A. S. 1988. Transient association of newly synthesized unfolded proteins with the heat-shock GroEL protein. *Nature* 336:254–57
7. Bukau, B., Walker, G. C. 1989. Cellular defects caused by deletion of the *Escherichia coli dnaK* gene indicate roles for heat shock protein in normal metabolism. *J. Bacteriol.* 171:2337–46
8. Bukau, B., Walker, G. C. 1990. Mutations altering heat shock specific subunit of RNA polymerase suppress major cellular defects of *E. coli* mutants lacking the DnaK chaperone. *EMBO J.* 9:4027–36
9. Calendar, R., Erickson, J. W., Halling, C., Nolte, A. 1988. Deletion and insertion mutations in the *rpoH* gene of *Escherichia coli* that produce functional σ^{32}. *J. Bacteriol.* 170:3479–84
10. Cooper, S., Ruettinger, T. 1975. A temperature sensitive nonsense mutation affecting the synthesis of a major protein of *Escherichia coli* K12. *Mol. Gen. Genet.* 139:167–76
11. Cowing, D. W., Bardwell, J. C. A., Craig, E. A., Woolford, C., Hendrix, R. W., Gross, C. A. 1985. Consensus sequence for *Escherichia coli* heat shock gene promoters. *Proc. Natl. Acad. Sci. USA* 82:2679–83
12. Craig, E. A., Gross, C. A. 1991. Is hsp70 the cellular thermometer? *TIBS* 16:135–40
13. Crickmore, N., Salmond, G. P. C. 1986. The *Escherichia coli* heat shock regulatory gene is immediately downstream of a cell division operon: the *fam* mutation is allelic with *rpoH*. *Mol. Gen. Genet.* 205:535–39
14. DiDomenico, B. J., Bugaisky, G. E., Lindquist, S. 1982. The heat shock response is self-regulated at both the transcriptional and posttranscriptional levels. *Cell* 31:593–603
15. Donnelly, C., Walker, G. C. 1989. *groE* mutants of *Escherichia coli* are defective in *umuDC*-dependent UV mutagenesis. *J. Bacteriol.* 171:6117–25
16. Drahos, D. J., Hendrix, R. W. 1982. Effect of bacteriophage lambda infection on synthesis of *groE* protein and other *Escherichia coli* proteins. *J. Bacteriol.* 149:1050–63
17. Ellis, R. J., van der Vies, S. M. 1991. Molecular chaperones. *Annu. Rev. Biochem.* 60:321–47
18. Erickson, J. W., Gross, C. A. 1989. Identification of the σ^E subunit of *Escherichia coli* RNA polymerase: a second alternate σ factor involved in high-temperature gene expression. *Genes Dev.* 3:1462–71
19. Erickson, J. W., Vaughn, V., Walter, W. A., Neidhardt, F. C., Gross, C.

A. 1987. Regulation of the promoters and transcripts of *rpoH*, the *Escherichia coli* heat shock regulaory gene. *Genes Dev.* 1:419–32

20. Faxen, M., Plumbridge, J., Isaksson, L. A. 1991. Codon choice and potential complementarity between mRNA downstream of the initiation codon and bases 1471–1480 in 16S ribosomal RNA affects expression of *glnS*. *Nucleic Acids Res.* 19:5247–51
21. Fujita, N., Ishihama, A. 1987. Heat-shock induction of RNA polymerase sigma-32 synthesis in *Escherichia coli:* transcriptional control and multiple promoter system. *Mol. Gen. Genet.* 210: 10–15
22. Fujita, N., Ishihama, A., Nagasawa, Y., Ueda, S. 1987. RNA polymerase sigma-related proteins in *Escherichia coli:* detection by antibodies against a synthetic peptide. *Mol. Gen. Genet.* 210:5–9
23. Fujita, N., Nomura, T., Ishihama, A. 1987. Promoter selectivity of *Escherichia coli* RNA polymerase: purification and properties of holoenzyme containing the heat-shock σ subunit. *J. Biol. Chem.* 262:1855–59
24. Gaitanaris, G. A., Papavassiliou, A. G., Rubock, P., Silverstein, S. J., Gottesman, M. E. 1990. Renaturation of denatured λ repressor requires heat shock proteins. *Cell* 61:1013–20
25. Gamer, J., Bujard, H., Bukau, B. 1992. Physical interaction between heat shock proteins DnaK, DnaJ, and GrpE and the bacterial heat shock transcription factor σ^{32}. *Cell* 69:833–42
26. Garvin, L. D., Hardies, S. C. 1989. Nucleotide sequence for the *htpR* gene from *Citrobacter freundii*. *Nucleic Acids Res.* 17:4889
27. Georgopoulos, C., Ang, D., Liberek, K., Zylicz, M. 1990. Properties of the *Escherichia coli* heat shock proteins and their role in bacteriophage λ growth. See Ref. 62, pp. 191–221
28. Georgopoulos, C., Ang, D., Maddock, A., Raina, S., Lipinska, B., Zylicz, M. 1990. Heat shock response of *Escherichia coli*. In *The Bacterial Chromosome,* ed. K. Drlica, M. Riley, pp. 405–19. Washington, DC: Am. Soc. Microbiol.
29. Goff, S. A., Casson, L. P., Goldberg, A. L. 1984. Heat shock regulatory gene *htpR* influences rates of protein degradation and expression of the *lon* gene in *Escherichia coli*. *Proc. Natl. Acad. Sci. USA* 81:6647–51
30. Goff, S. A., Goldberg, A. L. 1985. Production of abnormal proteins in *E. coli* stimulates transcription of *lon* and other heat shock genes. *Cell* 41:587–95
31. Gross, C. A., Straus, D. B., Erickson, J. W., Yura, T. 1990. The function and regulation of heat shock proteins in *Escherichia coli*. See Ref. 62, pp. 167–89
32. Grossman, A. D., Erickson, J. W., Gross, C. A. 1984. The htpR gene product of E. coli is a sigma factor for heat-shock promoters. *Cell* 38:383–90
33. Grossman, A. D., Straus, D. B., Walter, W. A., Gross, C. A. 1987. σ^{32} synthesis can regulate the synthesis of heat shock proteins in *Escherichia coli*. *Genes Dev.* 1:179–84
34. Grossman, A. D., Taylor, W. E., Burton, Z. F., Burgess, R. R., Gross, C. A. 1985. Stringent response in *Escherichia coli* induces expression of heat shock proteins. *J. Mol. Biol.* 186: 357–65
35. Herendeen, S. L., VanBogelen, R. A., Neidhardt, F. C. 1979. Levels of major proteins of *Escherichia coli* during growth at different temperatures. *J. Bacteriol.* 139:185–94
36. Jaeger, J. A., Turner, D. H., Zuker, M. 1989. Improved predictions of secondary structures for RNA. *Proc. Natl. Acad. Sci. USA* 86:7706–10
37. Jenkins, D. E., Auger, E. A., Matin, A. 1991. Role of RpoH, a heat shock regulator protein, in *Escherichia coli* carbon starvation protein synthesis and survival. *J. Bacteriol.* 173:1992–96
38. Johnson, W. C., Moran, C. P. Jr., Losick, R. 1983. Two RNA polymerase sigma factors from *Bacillus subtilis* discriminate between overlapping promoters for a developmentally regulated gene. *Nature* 302:800–4
39. Kamath-Loeb, A. S., Gross, C. A. 1991. Translational regulation of σ^{32} synthesis: requirement for an internal control element. *J. Bacteriol.* 173: 3904–6
40. Kanemori, M., Yura, T. 1991. Physiological functions of the GroE heat shock proteins: effects of decreased levels of GroE. *Annu. Meet. Mol. Biol. Soc. Jpn. Abstr.* p. 196
41. Keller, J. A., Simon, L. D. 1988. Divergent effects of a *dnaK* mutation on abnormal protein degradation in *Escherichia coli*. *Mol. Micorbiol.* 2:31–41
42. Kitagawa, M., Wada, C., Yoshioka, S., Yura, T. 1991. Expression of ClpB, an analog of the ATP-dependent pro-

tease regulatory subunit in *Escherichia coli,* is controlled by a heat shock σ factor (σ^{32}). *J. Bacteriol.* 173:4247–53

43. Kochan, J., Murialdo, H. 1982. Stimulation of *groE* synthesis in *Escherichia coli* by bacteriophage lambda infection. *J. Bacteriol.* 149:1166–70
44. Kroh, H. E., Simon, L. D. 1990. The ClpP component of Clp protease is the σ^{32}-dependent heat shock protein F21.5. *J. Bacteriol.* 172:6026–34
45. Krueger, J. H., Walker, G. C. 1984. *groEL* and *dnaK* genes of *Escherichia coli* are induced by UV irradiation and nalidixic acid in an *htpR*$^{+}$-dependent fashion. *Proc. Natl. Acad. Sci. USA* 81:1499–1503
46. Kusukawa, N., Yura, T. 1988. Heat shock protein GroE of *Escherichia coli:* key protective roles against thermal stress. *Genes Dev.* 2:874–82
47. Landick, R., Vaughn, V., Lau, E. T., VanBogelen, R. A., Erickson, J. W., Neidhardt, F. C. 1984. Nucleotide sequence of the heat shock regulatory gene of E. coli suggests its protein product may be a transcription factor. *Cell* 38:175–82
48. Langer, T., Lu, C., Echols, H., Flanagan, J., Hayer, M. K., Hartl, F. U. 1992. Successive action of DnaK, DnaJ and GroEL along the pathway of chaperone-mediated protein folding. *Nature* 356:683–89
49. LaRossa, R. A., Van Dyk, T. K. 1991. Physiological roles of the DnaK and GroE stress proteins: catalysts of protein folding or macromolecular sponges? *Mol. Microbiol.* 5:529–34
50. Lemaux, P. G., Herendeen, S. L., Bloch, P. L., Neidhardt, F. C. 1978. Transient rates of synthesis of individual polypeptides in E. coli following temperature shifts. *Cell* 13:427–34
51. Lesley, S. A., Burgess, R. R. 1989. Characterization of the *Escherichia coli* transcription factor σ^{70}: localization of a region involved in the interaction with core RNA polymerase. *Biochemistry* 28:7728–34
52. Lesley, S. A., Jovanovich, S. B., Tse-Dinh, Y.-C., Burgess, R. R. 1990. Identification of a heat shock promoter in the *topA* gene of *Escherichia coli. J. Bacteriol.* 172:6871–74
53. Lesley, S. A., Thompson, N. E., Burgess R. R. 1987. Studies of the role of the *Escherichia coli* heat shock regulatory protein σ^{32} by the use of monoclonal antibodies. *J. Biol. Chem.* 262:5404–7
54. Liberek, K., Galitski, T. P., Zylicz, M., Georgopoulos, C. 1992. The DnaK chaperone modulates the heat shock response of *Escherichia coli* by binding to the σ^{32} transcription factor. *Proc. Natl. Acad. Sci. USA* 89:3516–20
55. Lipinska, B., Fayet, O., Baird, L., Georgopoulos, C. 1989. Identification, characterization and mapping of the *Escherichia coli htrA* gene, whose product is essential for bacterial growth only at elevated temperatures. *J. Bacteriol.* 171:1574–84
56. Lipinska, B., King, J., Ang, D., Georgopoulos, C. 1988. Sequence analysis and transcriptional regulation of the *Escherichia coli grpE* gene, encoding a heat shock protein. *Nucleic Acids Res.* 16:7545–62
57. Lipinska, B., Sharma, S., Georgopoulos, C. 1988. Sequence analysis of the *htrA* gene of *Escherichia coli:* a σ^{32}-independent mechanism of heat-inducible transcription. *Nucleic Acids Res.* 16:10053–67
58. Lonetto, M., Gribskov, M., Gross, C. A. 1992. The σ^{70} family: sequence conservation and evolutionary relationships. *J. Bacteriol.* 174:3843–49
59. Maurizi, M. R., Clark, W. P., Katayama, Y., Rudikoff, S., Pumphrey, J., Bowers, B., Gottesman, S. 1990. Sequence and structure of ClpP, the proteolytic component of the ATP-dependent Clp protease of *Escherichia coli. J. Biol. Chem.* 265: 12536–45
60. McCarty, J. S., Walker, G. C. 1991. DnaK as a thermometer: threonine-199 is site of autophosphorylation and is critical for ATPase activity. *Proc. Natl. Acad. Sci. USA* 88:9513–17
61. Mizobata, T., Akiyama, Y., Ito, K., Yumoto, N., Kawata, Y. 1992. Effects of the chaperonin GroE on the refolding of tryptophanase from *Escherichia coli:* refolding is enhanced in the presence of ADP. *J. Biol. Chem.* 267:17773–79
62. Morimoto, R., Tissières, A., Georgopoulos, C., eds. 1990. *Stress Proteins in Biology and Medicine.* Cold Spring Harbor, NY: Cold Spring Harbor Lab. 450 pp.
63. Nagai, H., Yano, R., Erickson, J. W., Yura, T. 1990. Transcriptional regulation of the heat shock regulatory gene *rpoH* in *Escherichia coli:* involvement of a novel catabolite-sensitive promoter. *J. Bacteriol.* 172:2710–15
64. Nagai, H., Yuzawa, H., Yura, T. 1991. Interplay of two *cis*-acting mRNA regions in translational control of σ^{32}

synthesis during the heat shock response of *Escherichia coli. Proc. Natl. Acad. Sci. USA* 88:10515–19

65. Nagai, H., Yuzawa, H., Yura, T. 1991. Regulation of the heat shock response in *E. coli:* involvement of positive and negative *cis*-acting elements in translational control of σ^{32} synthesis. *Biochimie* 73:1473–79
66. Nagai, H., Yuzawa, H., Yura, T. 1992. *DnaK-mediated negative control of the heat shock response in* E. coli: *evidence for translational attenuation in* σ^{32} *synthesis*. Presented at Meeting on Translational Control, Cold Spring Harbor, N.Y.
67. Narberhaus, F., Bahl, H. 1992. Cloning, sequencing, and molecular analysis of the *groESL* operon of *Clostridium acetobutylicum. J. Bacteriol.* 174: 3282–89
68. Neidhardt, F. C., VanBogelen, R. A. 1981. Positive regulatory gene for temperature-controlled proteins in *Escherichia coli. Biochem. Biophys. Res. Commun.* 100:894–900
69. Neidhardt, F. C., VanBogelen, R. A. 1987. Heat shock response. In Escherichia coli *and* Salmonella typhimurium: *Cellular and Molecular Biology,* ed. F. C. Neidhardt, J. L. Ingraham, K. B. Low, B. Magasanik, M. Schaechter, H. E. Umbarger, pp. 1334–45. Washington, DC: Am. Soc. Microbiol.
70. Neidhardt, F. C., VanBogelen, R. A., Lau, E. T. 1983. Molecular cloning and expression of a gene that controls the high-temperature regulon of *Escherichia coli. J. Bacteriol.* 153:597–603
71. Neidhardt, F. C., VanBogelen, R. A., Vaughn, V. 1984. The genetics and regulation of heat-shock proteins. *Annu. Rev. Genet.* 18:295–29
72. Neuwald, A. F., York, J. D., Majerus, P. W. 1991. Diverse proteins homologous to inositol monophosphatase. *FEBS Lett.* 294:16–18

72a. Newlands, J. T., Gaal, T., Mecsus, J., Gourse, R. L. 1993. Transcription of the *Escherichia coli rrnB* P1 promoter by the heat shock RNA polymerase ($E\sigma^{32}$) in vitro. *J. Bacteriol.* 175:661–68

73a. Obukowicz, M. G., Slater, N. R., Krivi, G. G. 1992. Enhanced heterologous gene expression in novel *rpoH* mutants of *Escherichia coli. Appl. Environ. Microbiol.* 58:1511–23

73. Osawa, T., Yura. T. 1981. Effects of reduced amount of RNA polymerase sigma factor on gene expression and growth of *Escherichia coli:* studies of the *rpoD40* (amber) mutation. *Mol. Gen. Genet.* 184:166–73
74. Paek, K.-H., Walker, G. C. 1987. *Escherichia coli dnaK* null mutants are inviable at high temperature. *J. Bacteriol.* 169:283–90
75. Parsell, D. A., Sauer, R. T. 1989. Induction of a heat shock–like response by unfolded protein in *Escherichia coli:* dependence on protein level not protein degradation. *Genes Dev.* 3:1226–32
76. Pelham, H. R. B. 1986. Speculations on the functions of the major heat shock and glucose-regulated proteins. *Cell* 46:959–61
77. Raina, S., Georgopoulos, C. 1990. A new *Escherichia coli* heat shock gene, *htrC,* whose product is essential for viability only at high temperatures. *J. Bacteriol.* 172:3417–26
78. Schultz, J. E., Latter, G. I., Matin, A. 1988. Differential regulation by cyclic AMP of starvation protein synthesis in *Escherichia coli. J. Bacteriol.* 170:3903–9
79. Shean, C. S., Gottesman, M. E. 1992. Translation of the prophage λ *cI* transcript. *Cell* 70:513–22
80. Shi, W., Zhou, Y., Wild, J., Adler, J., Gross, C. A. 1992. DnaK, DnaJ, and GrpE are required for flagellum synthesis in *Escherichia coli. J. Bacteriol.* 174:6256–63
81. Skelly, S., Coleman, T., Fu, C.-F., Brot, N., Weissbach, H. 1987. Correlation between the 32-kDa σ factor levels and in vitro expression of *Escherichia coli* heat shock genes. *Proc. Natl. Acad. Sci. USA* 84:8365–69
82. Skowyra, D., Georgopoulos, C., Zylicz, M. 1990. The E. coli dnaK gene product, the hsp70 homolog, can reactivate heat-inactivated RNA polymerase in an ATP hydrolysis-dependent manner. *Cell* 62:939–44
83. Sprengart, M. L., Fatscher, H. P., Fuchs, E. 1990. The initiation of translation in *E. coli:* apparent base pairing between the 16S rRNA and downstream sequences of the mRNA. *Nucleic Acids Res.* 18:1719–23
84. Squires, C. L., Pedersen, S., Ross, B. M., Squires, C. 1991. ClpB is the *Escherichia coli* heat shock protein F84.1. *J. Bacteriol.* 173:4254–62
85. Strauch, K. L., Johnson, K., Beckwith, J. 1989. Characterization of *degP,* a gene required for proteolysis in the cell envelope and essential for growth of *Escherichia coli* at high temperature. *J. Bacteriol.* 171:2689–96

86. Straus, D. B., Walter, W. A., Gross, C. A. 1987. The heat shock response of *E. coli* is regulated by changes in the concentration of σ^{32}. *Nature* 329: 348–51
87. Straus, D. B., Walter, W. A., Gross, C. A. 1988. *Escherichia coli* heat shock gene mutants are defective in proteolysis. *Genes Dev.* 2:1851–58
88. Straus, D. B., Walter, W. A., Gross, C. A. 1989. The activity of σ^{32} is reduced under conditions of excess heat shock protein production in *Escherichia coli. Genes Dev.* 3:2003–10
89. Straus, D., Walter, W., Gross, C. A. 1990. DnaK, DnaJ, and GrpE heat shock proteins negatively regulate heat shock gene expression by controlling the synthesis and stability of σ^{32}. *Genes Dev.* 4:2202–9
90. Taglicht, D., Padan, E., Oppenheim, A. B., Schuldiner, S. 1987. An alkaline shift induces the heat shock response in *Escherichia coli. J. Bacteriol.* 169: 885–87
91. Tatti, K. M., Moran C. P. Jr. 1985. Utilization of one promoter by two forms of RNA polymerase from *Bacillus subtilis. Nature* 314: 190–92
92. Taura, T., Kusukawa, N., Yura, T., Ito, K. 1989. Transient shutoff of *Escherichia coli* heat shock protein synthesis upon temperature shift down. *Biochem. Biophys. Res. Commun.* 163: 438–43
93. Taylor, W. E., Straus, D. B., Grossman, A. D., Burton, Z. F., Gross, C. A., Burgess, R. 1984. Transcription from a heat-inducible promoter causes heat shock regulation of the sigma subunit of E. coli RNA polymerase. *Cell* 38:371–81
94. Tilly, K., Erickson, J., Sharma, S., Georgopoulos, C. 1986. Heat shock regulatory gene *rpoH* mRNA level increases after heat shock in *Escherichia coli. J. Bacteriol.* 168:1155–58
95. Tilly, K., McKittrick, N., Zylicz, M., Georgopoulos, C. 1983. The dnaK protein modulates the heat-shock response of Escherichia coli. *Cell* 34:641–46
96. Tilly, K., Spence, J., Georgopoulos, C. 1989. Modulation of stability of the *Escherichia coli* heat shock regulatory factor σ^{32}. *J. Bacteriol.* 171: 1585–89
97. Tobe, T., Ito, K., Yura, T. 1984. Isolation and physical mapping of temperature-sensitive mutants defective in heat-shock induction of proteins in *Escherichia coli. Mol. Gen. Genet.* 195: 10–16
98. Travers, A. A. 1982. The heat-shock phenomenon in bacteria—a protection against DNA relaxation? In *Heat Shock from Bacteria to Man*, ed. M. J. Schlesinger, M. Ashburner, A. Tissieres, pp. 127–30. Cold Spring Harbor, NY: Cold Spring Harbor Lab. 440 pp.
99. Tsuchido, T., VanBogelen, R. A., Neidhardt, F. C. 1986. Heat shock response in *Escherichia coli* influences cell division. *Proc. Natl. Acad. Sci. USA* 83:6959–63
100. VanBogelen, R. A., Acton, M. A., Neidhardt, F. C. 1987. Induction of the heat shock regulon does not produce thermotolerance in *Escherichia coli. Genes Dev.* 1:525–31
101. VanBogelen, R. A., Neidhardt, F. C. 1990. Ribosomes as sensors of heat and cold shock in *Escherichia coli. Proc. Natl. Acad. Sci. USA* 87:5589–93
102. Van Dyk, T. K., Gatenby, A. A., LaRossa, R. A. 1989. Demonstration by genetic suppression of interaction of GroE products with many proteins. *Nature* 342:451–53
103. Wada, C., Imai, M., Yura, T. 1987. Host control of plasmid replication: requirement for the σ factor σ^{32} in transcription of mini-F replication initiator gene. *Proc. Natl. Acad. Sci. USA* 84:8849–53
104. Wang, P.-Z., Doi, R. H. 1984. Overlapping promoters transcribed by *Bacillus subtilis* σ^{55} and σ^{37} RNA polymerase holoenzymes during growth and stationary phases. *J. Biol. Chem.* 259: 8619–25
105. Wang, Q., Kaguni, J. M. 1989. DnaA protein regulates transcription of the *rpoH* gene of *Escherichia coli. J. Biol. Chem.* 264:7338–44
106. Wang, Q., Kaguni, J. M. 1989. A novel sigma factor is involved in expression of the *rpoH* gene of *Escherichia coli. J. Bacteriol.* 171: 4248–53
107. Welch, W. J., Georgopoulos, C. 1993. Function and regulation of heat shock proteins. *Annu. Rev. Cell Biol.* In press
108. Wetzstein, M., Völker, U., Dedio, J., Löbau, S., Zuber, U., et al. 1992. Cloning, sequencing, and molecular analysis of the *dnaK* locus from *Bacillus subtilis. J. Bacteriol.* 174:3300–10
109. Wild, J., Kamath-Loeb, A., Ziegelhoffer, E., Lonetto, M., Kawasaki, Y., Gross, C. A. 1992. Partial loss of function mutations in DnaK, the *Es-*

cherichia coli homologue of the 70-kDa heat shock proteins, affect highly conserved amino acids implicated in ATP binding and hydrolysis. *Proc. Natl. Acad. Sci. USA* 89:7139–43
110. Yamamori, T., Ito, K., Nakamura, Y., Yura, T. 1978. Transient regulation of protein synthesis in *Escherichia coli* upon shift-up of growth temperature. *J. Bacteriol.* 134:1133–40
111. Yamamori, T., Yura, T. 1980. Temperature-induced synthesis of specific proteins in *Escherichia coli:* evidence for transcriptional control. *J. Bacteriol.* 142:843–51
112. Yamamori, T., Yura, T. 1982. Genetic control of heat-shock protein synthesis and its bearing on growth and thermal resistance in *Escherichia coli* K-12. *Proc. Natl. Acad. Sci. USA* 79:860–64
113. Yano, R., Nagai, H., Shiba, K., Yura, T. 1990. A mutation that enhances synthesis of σ^{32} and suppresses temperature-sensitive growth of the *rpoH15* mutant of *Escherichia coli. J. Bacteriol.* 172:2124–30
114. Yura, T., Tobe, T., Ito, K., Osawa, T. 1984. Heat shock regulatory gene (*htpR*) of *Escherichia coli* is required for growth at high temperature but is dispensable at low temperature. *Proc. Natl. Acad. Sci. USA* 81:6803–7
115. Yura, T., Yamamori, T., Osawa, T. 1983. Transcriptional control of heat shock-inducible operons in *Escherichia coli.* In *Microbiology—1983,* ed. D. Schlessinger, pp. 14–17. Washington, DC: Am. Soc. Microbiol.
116. Yuzawa, H., Nagai, H., Mori, H., Yura, T. 1991. *Thermoregulation of σ^{32} synthesis mediated by alternative mRNA secondary structures: a primary step of the heat shock response in* Escherichia coli. Presented at Meeting on Translational Control, Cold Spring Harbor, N.Y.
117. Zeilstra-Ryalls, J., Fayet, O., Georgopoulos, C. 1991. The universally conserved GroE (Hsp60) chaperonins. *Annu. Rev. Microbiol.* 45: 301–25
118. Zhou, Y.-N., Kusukawa, N., Erickson, J. W., Gross, C. A., Yura, T. 1988. Isolation and characterization of *Escherichia coli* mutants that lack the heat shock sigma factor σ^{32}. *J. Bacteriol.* 170:3640–49
119. Zhou, Y.-N., Walter, W. A., Gross, C. A. 1992. A mutant σ^{32} with a small deletion in conserved region 3 of σ has reduced affinity for core RNA polymerase. *J. Bacteriol.* 174:5005–12
120. Zylicz, M., Yamamoto, T., McKittrick, N., Sell, S., Georgopoulos, C. 1985. Purification and properties of the DnaJ replication protein of *Escherichia coli. J. Biol. Chem.* 260:7591–98